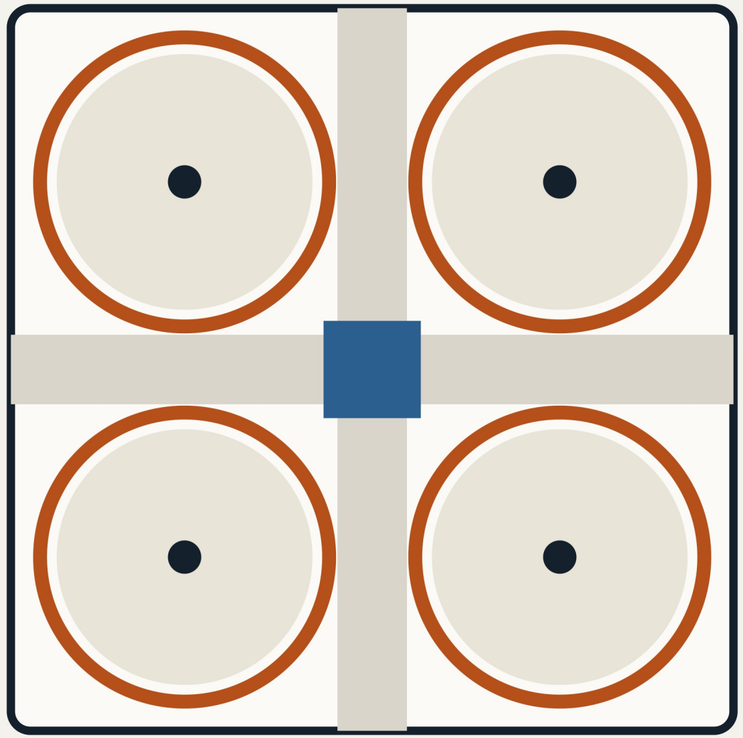
PROJECT 8 · TEAM 4 · EC4.402 · DESIGN REVIEW

Enclosed ducted-fan quadrotor

Four rotors inside a closed box: sized, specified and planned for autonomous flight.

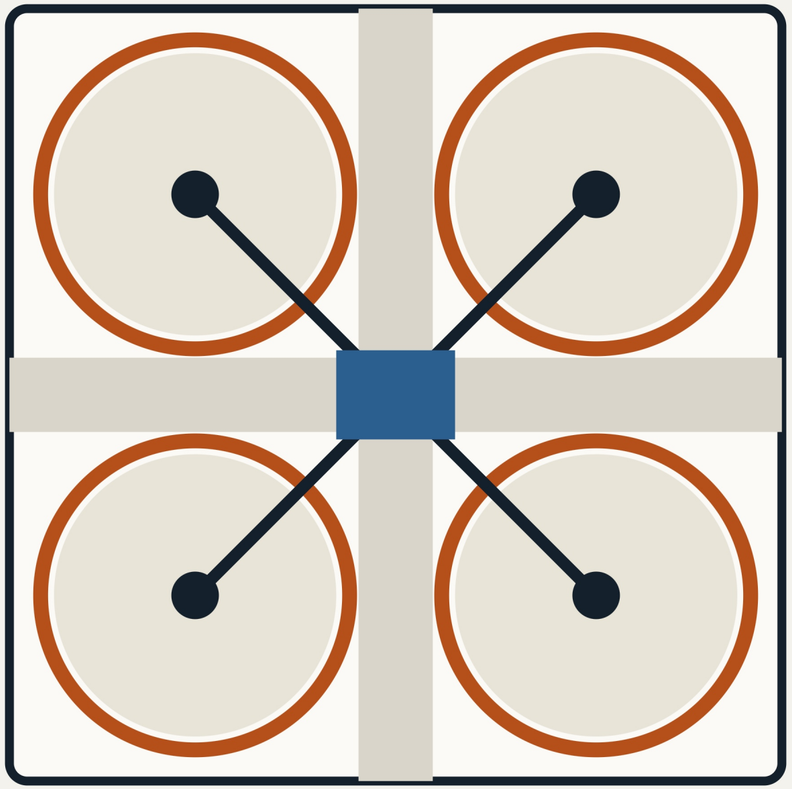


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What we are building, and how each goal gets checked

Goal	What we will show	Evidence
Fully enclosed propulsion	Four ducted fans inside a closed box, grilles on intake and exhaust, no exposed rotating parts	CAD and inspection of the built aircraft
Stable enclosed flight	Hover and position hold with the full enclosure fitted	Flight log and observed test
Autonomous mission	Auto take-off, ≥4 waypoints, auto landing, enclosure intact	Mission log and post-flight check
Safe operation	RC-loss and low-battery failsafes, geofence	Ground checks and a controlled test
Box form factor	Cuboid envelope, internal payload bay, nothing protruding below in flight	CAD and inspection of the built aircraft
Sound engineering	Sizing loop, then measured vs predicted thrust and endurance	Thrust-stand data and the report

Four ducts in a square box, payload in the middle



- **Box:** four printed PLA quarters that fit a 256 mm print bed
- **Ducts:** PETG bellmouth rings, 1.5 mm tip gap, grid grilles above and below
- **Frame:** carbon true-X inside the box, carrying motors at the duct centres
- **Cross spine:** 45 mm (5") / 55 mm (7") wide, ties the quarters together
- **Payload bay:** 60 × 40 × 30 mm at the centre of gravity, camera window; GPS mast on top

	5-inch	7-inch
Box width	353 mm	~475 mm
Duct outer diameter	~146 mm	~202 mm

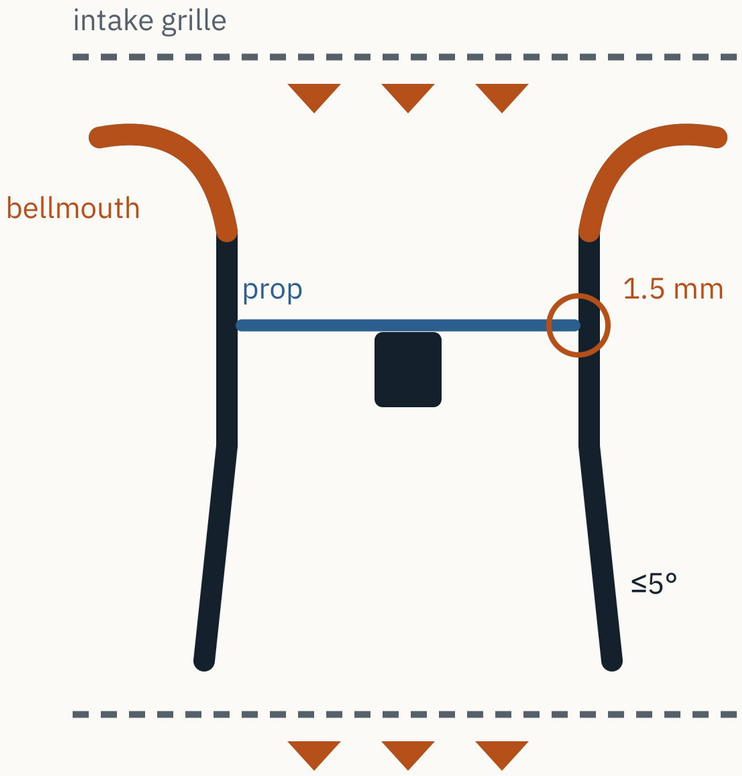
$$\text{Box} \approx 2a + d_o + 2 \times 6 = 195 + 146 + 12 = 353 \text{ mm}$$

$$\text{Duct gap } 2a - d_o = 48.8 \text{ mm} > 45 \text{ mm spine} \checkmark$$

$$\text{Motor to motor } 2\sqrt{2}\cdot a = 2\sqrt{2} \times 97.5 = 276 \text{ mm}$$

03 · DUCT

Each duct: rounded inlet, tight gap, short diffuser



Bellmouth inlet

A rounded lip lets air turn into the duct smoothly. A sharp edge separates the flow and loses thrust.

1.5 mm tip gap

Small gap cuts tip-vortex loss. Arms flex ~0.5 mm (5") / ~1.3 mm (7") at full thrust; 1.0 mm only on the rigid stand.

Short diffuser, ≤5°

Exit widens gently to recover pressure without the flow peeling off the wall.

Grid grilles

1 mm bars, 12 mm pitch: finger-safe, ~16% blocked. Raised 10–15 mm above the lip so corner air turns into the bore.

$$d_b = D + 2\delta = 127 + 2(1.5) = 130 \text{ mm}$$

$$d_o = d_b + 2(r_{lip} + t) = 130 + 2(6.5 + 1.6) = 146.2 \text{ mm}$$

$$\beta = 1 - ((s - b)/s)^2 = 1 - (11/12)^2 = 0.16$$

$$\text{Intake } (1 - \beta)\cdot\text{area} / (\pi D^2/4) = 1.42 \text{ at grille} \cdot 0.88 \text{ at bore}$$

04 · SYSTEM ARCHITECTURE

Sense, decide, act: ArduPilot (Copter) at the centre

SENSE & LINK

GPS + compass
Matek M10Q, SBAS

RC receiver
Lab set, manual override

Telemetry radio
433 / 865 MHz ↔ Mission Planner

→

DECIDE

SpeedyBee F405 V4 · ArduPilot Copter

- EKF fuses IMU, barometer and GPS
- Runs the mission: take-off, waypoints, landing
- Failsafes: RC loss, low battery, geofence
- Logs every flight to SD

→

ACT

55 A 4-in-1 ESC
→ 4 motors in ducts

Leg servos (if fitted)
2 × MG90S, retract in flight

Payload camera
RunCam Thumb records the mission

POWER 6S LiPo → ESC → motors · ESC current sensor → FC for the battery failsafe · FC 5 V → avionics · separate 5 V 3 A UBEC → leg servos, so a stalled servo can’t brown out the FC

05 · PROP SIZE

We start with 5-inch; 7-inch is the fallback

TRADE STUDY · LEGS + 180 G PAYLOAD

	5-inch	6-inch	7-inch
Duct outer diameter	**146 mm**	174 mm	202 mm
Box side	**353 mm**	414 mm	475 mm
All-up mass	**1,905 g**	2,320 g	2,735 g
Battery, 6S	**2,600 mAh**	2,950 mAh	3,300 mAh
Thrust per motor	**1.41 kgf**	1.59 kgf	1.77 kgf
Thrust-to-weight	**2.97**	2.75	2.59
Hover power	**516 W**	577 W	632 W
Hover, 0 / 20% loss	**5.4 / 4.5 min**	5.4 / 4.6 min	5.6 / 4.6 min

Hover time barely changes: a bigger prop's efficiency is spent carrying a bigger box and pack. 5-inch has the most margin, lightest mass and smallest box. 6-inch interpolated from the full 5- and 7-inch BOMs.

WE SWITCH TO 7-INCH IF

≥1.19 kgf per motor (duct + grilles): 5-inch as designed

1.10–1.19 kgf: keep 5-inch, cut ~140 g (thinner grilles, lighter panels)

0.95–1.10 kgf: 5-inch only by accepting T/W ≈ 2.0–2.3; otherwise 7-inch

Below 0.95 kgf, or electronics won't fit the 353 mm box: 7-inch

7-inch is already sized (2807 1300KV, Gemfan 7040-3, 6S 3300) but needs ~300 mm legs or cradle. Its 1.77 kgf comes from a measured 2807 1300KV test on the 7040-3 at 24 V (2.48 kgf); Emax's 2.64 kgf listing gives no prop or voltage.

06 · LANDING GEAR

Skids first; retractable legs if the weight allows

BASELINE

Flush skids + ground cradle

TPU pads flush with the box bottom: no mechanism, lightest option, nothing to fail mid-mission.

Downsides: take-off and landing need a cradle ≥250 mm tall so the exhaust can escape, and auto-landing must hit it (~±1 m). No safe landing on open ground.

UPGRADE

Retractable legs

2 × MG90S on their own 5 V 3 A supply; ~240 mm legs with lateral feet, hinges offset so they fold past each other. Lands anywhere flat.

We fit them if the thrust stand gives ≥1.19 kgf per motor (T/W 2.5 with the legs' +80 g) and enclosed hover stays ≥5 min.

PAYLOAD RunCam Thumb (9.8 g) in a 60 × 40 × 30 mm bay records the mission. The TA suggested a payload of ~10% of all-up weight, so we add 170 g of ballast: 180 g total.

The sizing loop, step by step, at the converged 1,905 g

THE LOOP	THE CALCULATION
1 **Guess the mass** Start at a worst case	m_0 = 2.0 kg → after re-weighing, m = **1.905 kg**
2 **Work out hover need** Thrust per motor, then power by momentum theory, then current	$T_h = mg/4 = 1.905 \times 9.81 / 4 =$ **4.67 N** $P_i = T_h^{3/2} / \sqrt{(2\rho A)} =$ **60.8 W** → $P_e = P_i / (FM \cdot \eta) =$ **126.6 W** $I_h = (4P_e + P_{av}) / V = 516 / 22.2 =$ **23.3 A**
3 **Pick motor, prop, pack** Need T/W ≥2.5, ~5 min hover, pack able to supply full-throttle current	$T_{max} = 1.98 \times (1.09/1.225) \times (21.5/24)^2 =$ **1.41 kgf** $T/W = 4 \times 1.41 / 1.905 =$ **2.97** $t_h = 0.8 \cdot C / I_h = 0.8 \times 2.6$ Ah / 23.3 A = 0.089 h = **5.4 min** 4 × 40 A = 160 A ≤ 2.6 Ah × 65 C = **169 A** ✓
4 **Re-weigh** Sum the BOM; estimated parts get +15%	m = Σspec + 1.15·Σest + allow = 612 + 1.15 × 950 + 200 = **1,905 g**
5 **Converged?** Under 2% change: stop. Otherwise back to step 2	Round 2: Δm/m = −4.8% · Round 3: Δm/m = **0%** ✓ stop

Inputs: ρ = 1.09 kg/m³ (540 m, 30 °C) · A = $\pi D^2/4$ = 0.0127 m² · FM = 0.6 · η = 0.8 · P_{av} = 10 W · V = 22.2 V (21.5 V loaded)

Predicted weight, part by part

Group	What's in it	5-inch (g)	7-inch (g)
Battery	6S 2600 / 3300 mAh	400	520
Box structure	Panels, decks, spine, bay door	388	543
Ducts ×4	PETG bellmouth rings	230	460
Propulsion	Motors, leads, props	173	240
Grilles ×8	Intake and exhaust	156	299
Frame	Carbon true-X	92	155
Wiring, fasteners	XT90, inserts, bolts	92	104
Avionics + radios	FC, ESC, GPS, RC, telemetry	87	87
Skids	TPU pads	28	41
Payload	RunCam Thumb + 170 g ballast	180	180
All-up, skids		**1,825**	**2,629**

5-INCH BASELINE

1,825 g

With legs: 1,905 g (+80 g)
7-inch: 2,629 g · with legs 2,735 g

How sure are we?

```
m = Σspec + 1.15·Σest + allow
= 585 + 1.15 × 904 + 200
= 1,825 g ✓
```

57% is estimated (printed parts, frame, wiring), so we weigh a printed duct and grille first.

Key components

Component	Choice (5" / 7")
Flight controller	SpeedyBee F405 V4, ArduPilot
ESC	SpeedyBee BLS 55 A 4-in-1
GPS + compass	Matek M10Q-5883
RC + telemetry	Lab sets, 433 / 865 MHz
Motors ×4	ECO II 2207 1700KV / 2807 1300KV
Props, tri-blade	5×4.3 / Gemfan 7040-3
Battery	6S 2600 ≥65C / 3300 ≥60C, XT90
Power wiring	12 AWG, ≥50 V low-ESR cap
Payload camera	RunCam Thumb

Component	Choice (5" / 7")
Frame	Carbon true-X, ~276 / ~369 mm
Enclosure	PLA: 4 quarters, spine and decks
Ducts ×4	PETG (PETG-CF if possible) bellmouths
Grilles ×8	PLA grid, 1 mm bars, 12 mm pitch
Landing gear A	Legs with PA-CF hinges, 2 × MG90S + UBEC
Landing gear B	TPU skids + open-frame cradle
Charger	6S balance, ≥100 W
Thrust stand	Lab stand, or load cell + HX711
Filament	PLA box · PETG ducts · PA-CF mounts

Fly it bare first, then enclose it

STAGE 0

Thrust stand

One motor in a printed duct with both grilles, on the real 6S pack. Sweep tip gap 1.0 / 1.5 / 2.0 mm.

Gate: ≥1.14 kgf per motor (≥1.19 with legs) for T/W 2.5; below that, the rule on slide 6

STAGE 1

Bare quad

Frame and avionics, no enclosure. Tune, hover, position hold, failsafe checks.

Gate: stable hover and position hold, CG within 5 mm, vibration under 15 m/s², clean logs

STAGE 2

Enclosed

Add ducts, box and grilles; retune. Measure hover current against the model.

Gate: stable enclosed hover, enclosure loss measured

STAGE 3

Mission

Auto take-off, ≥4 waypoints, auto landing. Three full rehearsals.

Gate: lands with ≥25% battery, enclosure intact

If enclosed thrust comes in short, we tune the propulsion (prop pitch, KV, tip gap) with a flying baseline already in hand.

11 · DELIVERABLES

Ten things we commit to deliver

01 Four ducted fans inside a closed box

02 Sizing loop, checked against measurements

03 CAD and the fabricated enclosure

04 Flying prototype with the FC integrated

05 Stable flight with the enclosure fitted

06 Autonomous mission: take-off, ≥4 waypoints, landing, ≥25% battery left

07 Landing gear: skids, legs if weight allows

08 Payload bay carrying 180 g, including a camera recording the mission

09 Failsafes: return-to-launch, low-battery land, geofence

10 Flight logs and final report

TARGETS T/W ≥2.5 · hover ≥4.5 min · waypoints within 2 m · 3 rehearsals · daylight, wind under 15 km/h

Where our design choices come from

SIZING MODEL	J. G. Leishman, *Principles of Helicopter Aerodynamics*, 2nd ed., Cambridge University Press, 2006: momentum theory, hover power and figure of merit
DUCT GEOMETRY	J. L. Pereira, *Hover and Wind-Tunnel Testing of Shrouded Rotors for Improved Micro Air Vehicle Design*, PhD thesis, Univ. of Maryland, 2008: tip clearance, inlet lip and diffuser effects drum.lib.umd.edu/handle/1903/8752
PRINTED DUCTED PLATFORM	A. D. Ryals et al., “A Modular 3D-Printed Ducted-Fan Platform for Advanced Autonomy Research: From Design to Flight Test,” *Drones* 10(3), 165, 2026 doi.org/10.3390/drones10030165
AUTOPILOT	ArduPilot Copter docs: failsafes (radio, battery, geofence) and retractable landing gear ardupilot.org/copter/docs/failsafe-landing-page.html
COMPONENT DATA	Manufacturer datasheets: Emax ECO II, SpeedyBee, Matek, Gemfan, RunCam